CycleMate

Helping IVF patients finish the treatment they started

A web-based companion for the IVF cycle · patient app + coordinator view · EN / UZ / RU

AI in Medicine & Biostatistics
Mid-term project · My (Digital) Health App
Hans · October 2026

1 · THE HEALTH PROBLEM

Infertility affects about **1 in 6 adults** worldwide (WHO, 2023). IVF is effective — but only when a patient completes the cycle.

The gap: up to **1 in 5 patients drop out** of IVF before finishing, and the leading reason is not medical failure. It is the **burden of treatment itself**. Most who stop never return.

17%
of couples drop out mid-treatment
Verberg 2008, n=384

19–28%
cite physical / psychological burden as the reason
Gameiro 2012 (n=21,453); Verberg 2008

18–27%
show clinical anxiety or depression during IVF
China 2025, n=2,465

Anxiety peaks at embryo transfer, not at the start (E-Freeze RCT, UK). The hardest days come late in the cycle.

WHAT "BURDEN" ACTUALLY IS

From clinic experience in Tashkent and the literature, four things wear patients down:

- Uncertainty — "what happens next?"
- Daily load — visits, pills, timing
- Feeling alone between visits
- Nowhere to ask a small question

TARGET USERS

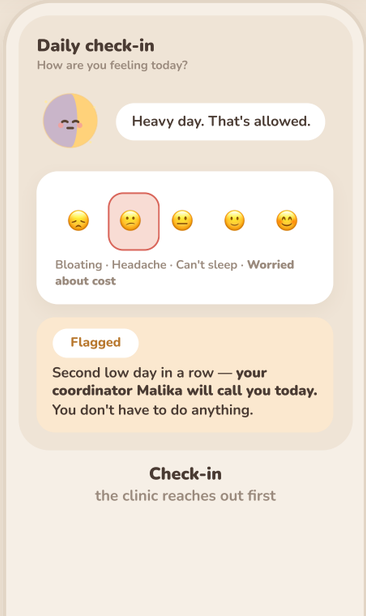
- **Patients** going through an IVF cycle — primary user. Many live far from Tashkent and cannot visit daily.
- **Clinic coordinators** — see who needs a call first, instead of waiting for patients to reach out.
- **Doctors** — receive only the questions that need a medical answer.

2 · THE APP — ONE BURDEN, ONE FEATURE

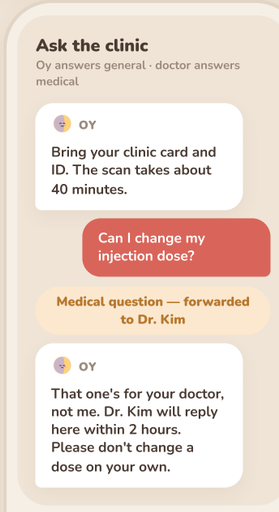
Uncertainty "what happens next?"	**Cycle timeline** — today's day, next step, estimated dates; shown as a moon that waxes with the cycle
Daily load	**Today's plan** — never more than 3 tasks; clinic visits, home medication, trigger-shot countdown
Feeling alone	**10-second check-in** — mood 1–5; two low days in a row and the coordinator calls *first*
Nowhere to ask	**Ask the clinic** — Oy answers general questions; anything medical is forwarded to the doctor
Cost worry	**Running total** — paid so far and estimated remaining; no surprises



Home
where am I in the cycle?



Check-in
the clinic reaches out first



Ask
AI knows what it must not answer

HOW IT WORKS

Patient
web app · any phone
phone-number login

⇄

Clinic server
encrypted · on-premise
consent-scoped sharing

⇄

Coordinator & doctor
call list · doctor queue

AI 1 · QUESTION ROUTER
Classifies each message as *general* (Oy answers from the clinic FAQ) or *medical* (forwarded to the doctor). Unknown → forwarded, never guessed.

AI 2 · CALL-LIST RANKING
Orders the coordinator's list from mood trend, missed tasks and unanswered questions. It changes who gets called first — never the treatment.

The AI never diagnoses, never changes a dose, never interprets a result. Every clinical decision stays with the medical team.

MEET OY — WHY A MOON

A lunar month and an IVF cycle are both about 28 days. Oy waxes through stimulation, is full at embryo transfer, and wanes toward the pregnancy test. One glance at the moon tells a patient where she is — no calendar, no reading.



Calm by design. Medical apps often use alerts, red badges and countdowns that raise anxiety. Oy replaces them with a soft, familiar companion who reacts to how the patient feels — so the app lowers fear instead of adding to it.

A moon, not an embryo. Oy stands for the *cycle*, not the outcome. Patients are never asked to bond with an embryo; if a cycle fails, the moon simply begins again.

reacts to check-ins, tasks and questions

3 · PRIVACY, SECURITY & ETHICS

PRIVACY & SECURITY
Login only with the phone number registered at the clinic (SMS code). Data encrypted in transit and at rest, stored on the clinic's own server. Patient chooses what to share — partner sees schedule only, never mood or messages. Deleted 12 months after the cycle, or on request.

ETHICS
AI gives general information only; medical questions go to a human. When unsure, it forwards rather than guesses. Flags prioritise a coordinator's calls — they never label or diagnose a patient. The companion is designed to reduce fear, not to build attachment to an outcome.

EQUITY & ACCESS

- **Web-based, no app store** — runs on any low-end phone; reminders fall back to SMS when offline.
- **Built for patients outside Tashkent** — only 4 visits are essential; remote check-ins and partner clinics cover the rest.
- **EN / UZ / RU** — icons and one task per screen for low-literacy users.

FEEDBACK → CHANGE · IMPACT · NEXT

Clinic coordinator: "Patients who are struggling rarely message us — they just stop coming." → **Added the 2-low-days rule:** the coordinator is flagged and calls first, instead of waiting for the patient.

v1 → v3 (self-review): a plain timer-style UI felt clinical and cold → added Oy, warm palette, one task at a time.

PRIMARY	SECONDARY	CLINIC
cycle completion rate	check-in trend · time-to-contact	proactive calls made

Next: EMR integration for automatic schedules · partner-clinic network for regional patients · partner mode · pilot with 20 patients, 1 cycle.